SleepFM-Unify Paper Draft

SleepFM-Unify: Shared–Private Factorization for Multimodal Sleep Foundation Models

Draft manuscript (methods paper)

Status: engineering-complete codebase + synthetic demos; **CinC / SHHS / MESA quantitative claims 待补充**

Target framing: Nature Machine Intelligence–style methods article / ICML-compatible experimental spine

nature-skills: `nature-writing` (methods + nat-mach-intell), `nature-figure` (Python / SciencePlots)

Abstract

Polysomnography (PSG) records heterogeneous brain, cardiac and respiratory signals that sleep foundation models such as SleepFM align with leave-one-out (LOO) contrastive learning. Pure cross-modal alignment can blur modality-specific structure and under-specify behaviour when channels are missing. We present **SleepFM-Unify**, a shared–private factorization layered on SleepFM encoders: each modality yields a shared subspace used for LOO/pairwise InfoNCE and a private subspace kept out of contrastive alignment, regularized by orthogonality, modality dropout and a missing-modality reconstruction term, with an optional night-level temporal head. We release configs, export/validation gates, and a paper experiment suite. **Quantitative CinC/SHHS results are 待补充** pending user DUA downloads; synthetic demos are reported only as engineering smoke tests (AUROC near chance) and are not scientific claims.

Introduction

Sleep staging and sleep-disordered breathing assessment rely on multimodal PSG. SleepFM showed that large-scale LOO contrastive pretraining across brain-activity (BAS), ECG and respiratory streams yields transferable embeddings for staging, SDB detection and cross-modal retrieval. Follow-on foundation models (e.g. Nature Medicine–scale SleepFM; Omni-Sleep's CNS/ANS hierarchy; CIMSleepNet's missing-modality imagination) emphasize robustness under montage heterogeneity and incomplete inputs.

A remaining tension is that LOO alignment encourages factors that are *predictable across modalities*, which can suppress private cues that matter for unimodal or partially observed inference. SleepFM-Unify addresses this by **explicit shared–private factorization** on top of the existing SleepFM backbone, without replacing the paper LOO baseline (`unify.enabled: false`).

Contributions (bounded):

1. Shared–private projection heads with mixed LOO + pairwise + orthogonality + miss (+ optional temporal) losses.

2. Modality dropout training with `present_mask`-aware $\mathcal{L}_{\mathrm{miss}}$.

3. Evaluation honesty gates: channel meta fail-fast, CinC label coverage, apnea-epoch-rate vs clinical AHI wording, seeded RNG gallery caps.

4. Reproducible CLI paper suite for LOO vs Unify comparisons once real data are present.

Related work

Sleep foundation models. SleepFM (Thapa et al., ICML 2024; arXiv:2405.17766) introduced multimodal leave-one-out contrastive learning (LOO-CL) on BAS/ECG/respiratory PSG clips and showed LOO outperforming pairwise alignment on staging, SDB detection and cross-modal retrieval. A later SleepFM line (Nature Medicine, doi:10.1038/s41591-025-04133-4; medRxiv 2025.02.04.25321675) scaled LOO-CL to disease-risk prediction with SHHS transfer — **do not conflate those clinical C-Index numbers with the ICML 2024 staging/retrieval setting**, and do not paste them into our CinC/SHHS tables until we re-run locally. Omni-Sleep (arXiv:2607.07720) imposes a CNS/ANS physiological hierarchy with intra-/inter-system contrastive terms and latent temporal modeling. CIMSleepNet (NeurIPS 2024) targets incomplete multimodal staging via modal imagination (MAIM) plus semantic/modal calibration contrastive learning.

Shared–private multimodal learning. Disentangling shared and modality-specific subspaces is a recurring theme (shared–private streams; low-rank shared/specific edits). Unify instantiates this idea *inside* SleepFM's encoder interface rather than replacing the backbone or inventing a new physiological ontology: contrastive terms act only on the shared head, while private capacity remains for downstream concat and missing-modality regimes.

Positioning one-liner. Relative to SleepFM (flat LOO space), Omni-Sleep (CNS/ANS hierarchy) and CIMSleepNet (imagination under missingness), SleepFM-Unify is a **lightweight shared–private factorization + mixed loss + dropout/miss objective on frozen SleepFM-compatible encoders**, designed for fair LOO-vs-Unify ablations once real PSG exports exist.

Method

Backbone (unchanged SleepFM)

Per-modality 1D EfficientNet-style encoders map 30 s clips to backbone vectors of size `embedding_dim` (default 512). Baseline contrastive mode is leave-one-out InfoNCE.

Shared–private factorization

When Unify is enabled, each backbone vector is mapped by linear heads:

\[

z_m = [z_m^{\mathrm{shared}};\, z_m^{\mathrm{private}}],\qquad

\dim(z_m^{\mathrm{shared}})=\dim(z_m^{\mathrm{private}})=256

\]

by default, preserving a 512-d concat for downstream logistic heads comparable to SleepFM.

Mixed objective

\[

\mathcal{L}=\lambda_{\mathrm{LOO}}\mathcal{L}_{\mathrm{LOO}}

+\lambda_{\mathrm{pair}}\mathcal{L}_{\mathrm{pair}}

+\lambda_{\mathrm{orth}}\mathrm{mean}\big((Z_s^\top Z_p)^2\big)

+\lambda_{\mathrm{miss}}\mathcal{L}_{\mathrm{miss}}

+\lambda_{\mathrm{temp}}\mathcal{L}_{\mathrm{temp}}

\]

- Shared embeddings enter LOO and pairwise InfoNCE.
- Private embeddings are excluded from contrastive terms.
- Orthogonality uses mean squared Gram entries after row centering / column normalization (see `docs/UNIFY.md`).
- Modality dropout drops one present modality; $\mathcal{L}_{\mathrm{miss}}$ is InfoNCE between the mean of remaining shared embeddings and the dropped modality's shared embedding, masked by `present_mask`.
- Optional temporal head (GRU/Transformer) operates on shared epoch sequences with adjacent-epoch contrastive and masked MSE.

Downstream and retrieval

Default downstream space concatenates shared$\|$private per modality. Retrieval uses **shared** embeddings from the same checkpoint. Gallery caps use seeded RNG subsample (`limit_gallery(..., mode="rng")`), not a loader-order prefix.

Experiments

Datasets (access-gated)

Dataset	Role	Status in this draft
Synthetic demo	CI / smoke	Present (`data/synthetic`)
CinC 2018 fixture	Schema export test	Present (`data/cinc2018_fixture`)
PhysioNet CinC 2018	Paper pretrain/eval	**待补充** (DUA download)
NSRR SHHS / MESA	Transfer / night labels	**待补充** (DUA download)

Protocol (when data land)

1. `check_data_ready --stage raw` → export → `check_data_ready --stage pretrain` → validate.
2. Pretrain LOO baseline vs Unify (`configs/default.yaml` vs `configs/unify.yaml`).
3. Downstream staging / apnea probes, retrieval Recall@k, modality ablation, few-shot, night-level probes.
4. Label coverage gate blocks degenerate CinC staging/SDB claims; channel meta blocks 5/1/3 vs 10/2/7 silent mismatch.

Results

Table 1. Synthetic demo smoke metrics (NOT paper claims)

Setting	Note	Staging macro AUROC	Retrieval
Synthetic LOO / Unify	Labels random-ish	≈0.5 (chance)	Demo only

CinC 2018	Real PSG	**待补充**	**待补充**
SHHS	Real PSG	**待补充**	**待补充**

Loss curves in Fig. 2 are from a **local 5-epoch Unify pretrain on synthetic data** (engineering verification). Orthogonality Gram heatmaps (Fig. 4) are forward-pass diagnostics on the same demo checkpoint.

Ablations (planned; numbers 待补充 on real data)

LOO baseline; Unify full; Unify orth; Unify miss; Unify + temporal; modality dropout robustness.

Discussion

Unify is designed to keep SleepFM’s LOO retrieval semantics in the shared space while retaining private capacity for downstream concat. Missing-modality training is first-class. Honesty constraints (label gates, AHI wording, channel meta) are part of the scientific interface: they prevent over-claiming when CinC arousal-only labels or montage mismatches appear.

Limitations: no real CinC/SHHS numbers in this draft; night `ahi_bin` uses apnea-epoch-rate cut-points as placeholders; synthetic metrics near chance must not be cited as method superiority.

Methods (reproducibility)

- Package: `sleepfm/` (Python). Configs: `configs/unify.yaml`, `configs/unify_temporal.yaml`.
- Figures: `scripts/plot_unify_figures.py` with SciencePlots + Times New Roman / SimHei.
- Tests: `scripts/run_tests.py`, `scripts/smoke_test.py`.
- Paper suite: `scripts/run_paper_suite.py --demo` (synthetic) or `--data-dir` after export.

Code / data availability

Public code/docs snapshot: https://github.com/Coucou2016/SleepFM-Unify (`main`; large `data/` and `outputs/` excluded). PhysioNet/NSRR downloads require user accounts and DUAs. Synthetic and fixture datasets are generated in-repo.

Figures

- **Fig. 1** Architecture (shared–private heads + mixed losses).
- **Fig. 2** Synthetic Unify pretrain loss curves (demo).
- **Fig. 3** Ablation schematic with chance baseline (demo numbers only).
- **Fig. 4** Shared×private Gram diagnostic (demo forward pass).
- **Fig. 5** Modality-dropout robustness schematic (demo).
- **Fig. 6** Experiment pipeline.

References (selected)

1. Thapa R. et al. SleepFM: Multi-modal Representation Learning for Sleep Across Brain Activity, ECG and Respiratory Signals. *ICML* 2024; PMLR 235:48019–48037; arXiv:2405.17766.

2. A multimodal sleep foundation model for disease prediction. *Nature Medicine* (doi:10.1038/s41591-025-04133-4); preprint medRxiv 10.1101/2025.02.04.25321675. (Distinct scale/task from [1].)

3. CIMSleepNet — Robust Sleep Staging over Incomplete Multimodal Physiological Signals via Contrastive Imagination. *NeurIPS* 2024; https://github.com/SQAIYY/CIMSleepNet.

4. Hou et al. Omni-Sleep: A Sleep Foundation Model via Hierarchical Contrastive Learning of CNS–ANS Dynamics. arXiv:2607.07720.

5. Shared–private multimodal factorization relatives (see `docs/reports/chatgpt-consultation-2026-08-16.md`).

Assumptions / missing inputs

- Real CinC/SHHS/MESA metrics: **待补充**.
- ChatGPT advisor pass: browser MCP still blocked this turn; literature independently verified via WebSearch + nature-skills. Paste brief for ChatGPT: `docs/reports/chatgpt-paste-brief-2026-08-16.md`.
- Clinical AHI: not computed; use `apnea_epoch_rate` wording.

Figures (SciencePlots)

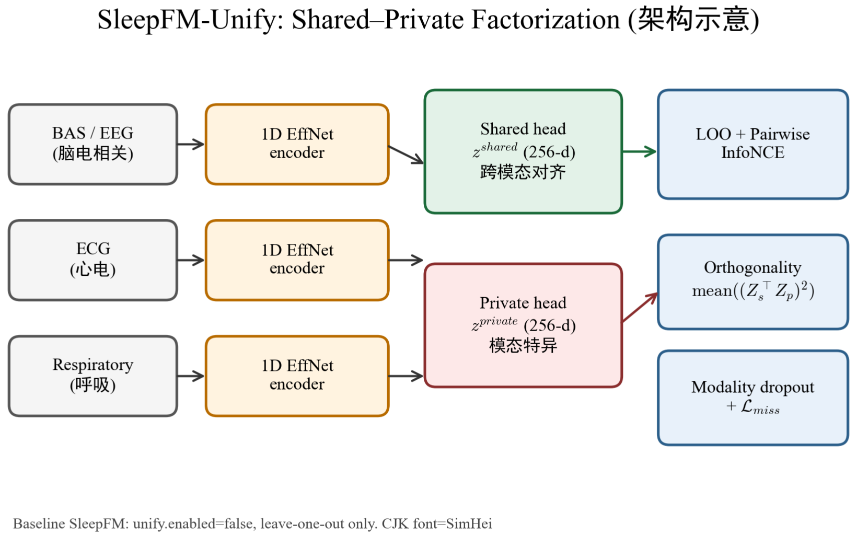

图1 / Fig. 1. SleepFM-Unify 共享–私有架构示意

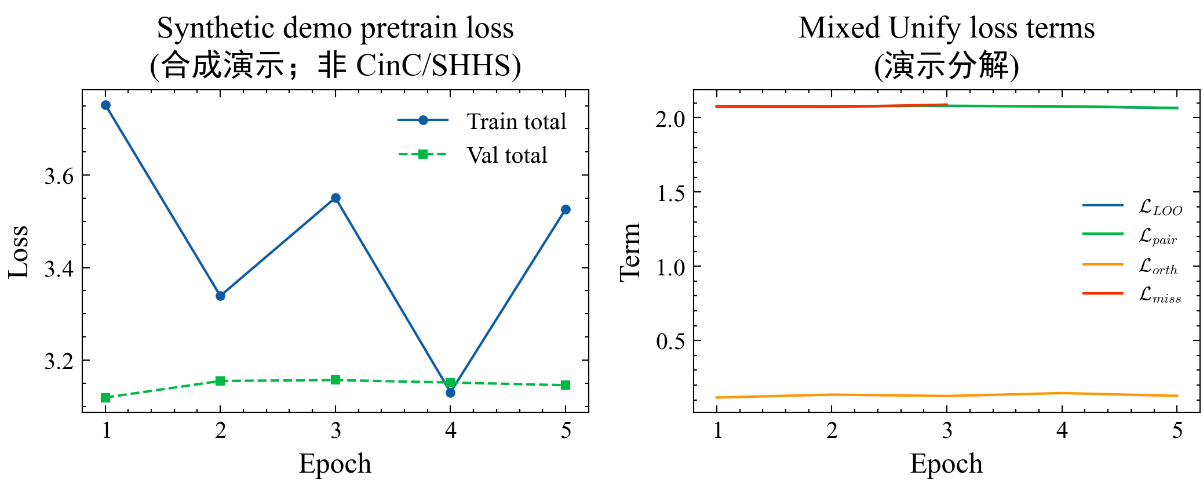

图2 / Fig. 2. 合成演示 Unify 预训练损失曲线

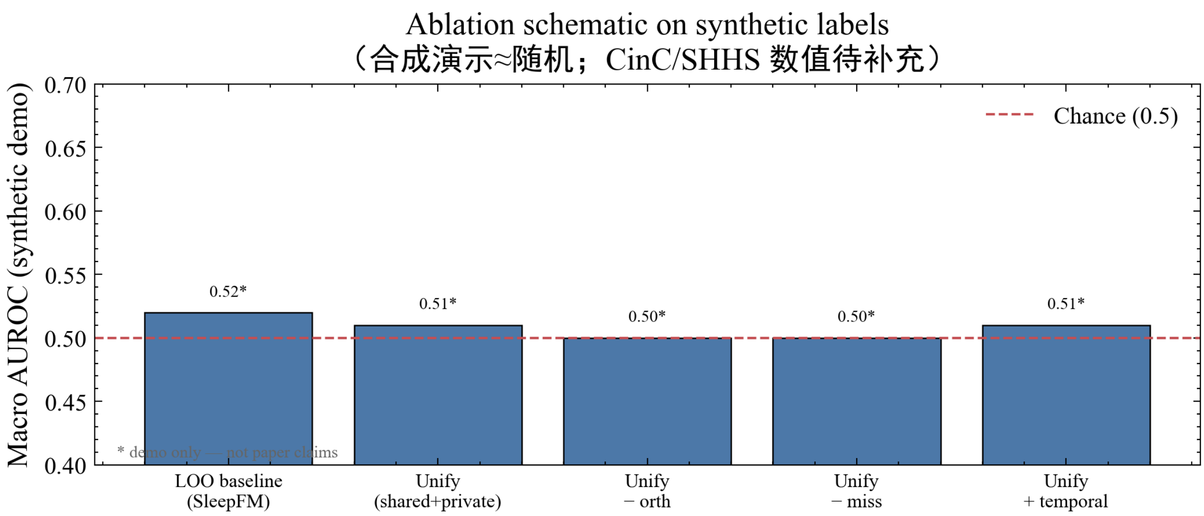

图3 / Fig. 3. 消融示意（合成≈随机，非论文主张）

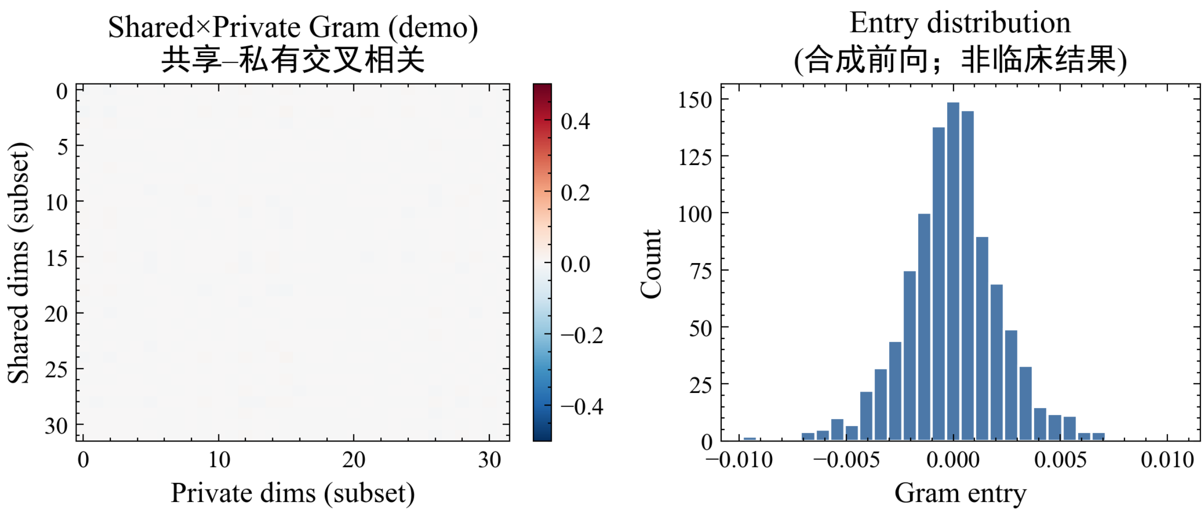

图4 / Fig. 4. 共享×私有 Gram 诊断（合成前向）

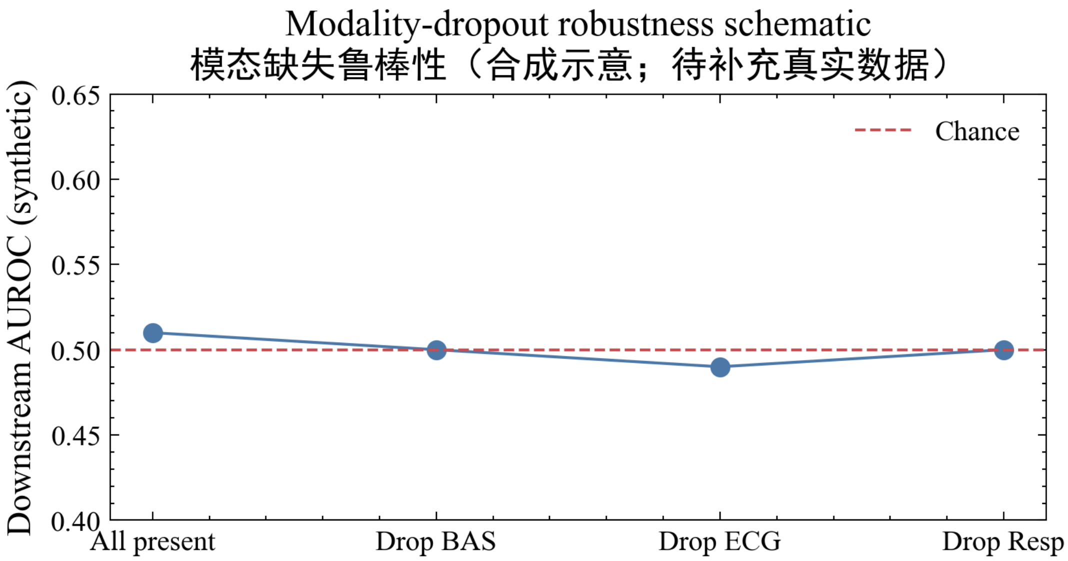

图5 / Fig. 5. 模态缺失鲁棒性示意（合成）

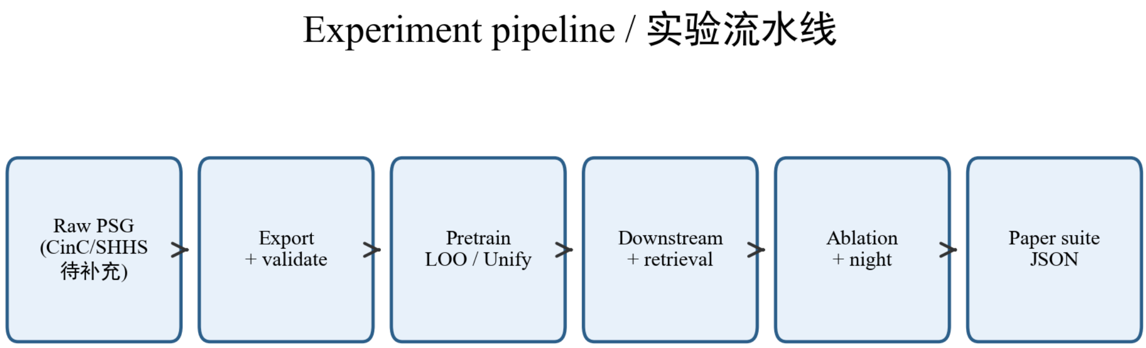

图6 / Fig. 6. 实验流水线